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Division: CS-1

Batch: C13

Course: EDS

Essential of Data Science Laboratory

Practical 1:

1.1.1 Calculate Momentum:

The screenshot displays the CODETANTRA online IDE interface. On the left, the problem statement for "1.1.1. Calculate Momentum" is shown, including the formula $p = m \times v$ and input/output specifications. The main editor shows a Python script that takes mass and velocity as input and prints the momentum. The right sidebar displays the test results, indicating that 2 out of 2 shown test cases and 2 out of 2 hidden test cases passed. The test cases show expected and actual outputs for two different inputs.

1.1.1. Calculate Momentum

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:

$$p = m \times v$$

where:

- m is the mass of the object (in kilograms).
- v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases

calculate...

```
1 m=float(input())
2 v=float(input())
3 p=m*v
4 print("%.2fkgm/s"%p)
```

Average time: 0.025 s (24.75 ms) | Maximum time: 0.034 s (34.00 ms)

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 (34 ms)

Expected output	Actual output
5.0	5.0
10.0	10.0
50.00kgm/s	50.00kgm/s

Test case 2 (20 ms)

Expected output	Actual output
2.3	2.3
4.8	4.8
11.04kgm/s	11.04kgm/s

Terminal | Test cases

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1.1.2 Conditional Calculations:

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1.1.2. Conditional Calculation Based on the Number of Digits

29.47

Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Constraints:

- $1 \leq n \leq 999$

Input Format:

- The input consists of a single integer n .

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases

condition...

Submit

```
1 import math
2 n= int(input())
3 if 1<= n <=9:
4     print(n*n)
5 elif 10 <= n <=99:
6     s= math.sqrt(n)
7     print("%.2f"%s)
8 elif 99 < n <= 999:
9     c= math.cbrt(n)
10    print("%.2f"%c)
11 else:
12    print("Invalid")
```

Average time

0.021 s

21.14 ms

Maximum time

0.032 s

32.00 ms

4 out of 4 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1

32 ms

Debug

Expected output

Actual output

9

81

Test case 2

18 ms

Test case 3

18 ms

Test case 4

26 ms

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1.1.3 Days Between two Dates:

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1.1.3. Days between Two Dates

11.42

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format $YYYY-MM-DD$.
- The second line contains the second date in the format $YYYY-MM-DD$.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases

noOfDay...

Submit

```
1 from datetime import date
2 date1 = input().strip()
3 date2 = input().strip()
4
5 d1 = date.fromisoformat(date1)
6 d2 = date.fromisoformat(date2)
7
8 diff = (d2-d1).days
9
10 print(diff)
11
```

Average time

0.016 s

16.00 ms

Maximum time

0.017 s

17.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

17 ms

Debug

Expected output

Actual output

2014-07-02

2014-07-02

2014-11-02

123

Test case 2

15 ms

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1.1.4 Reverse a Number:

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1.1.4. Reverse a Number13/26

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

The input is a single integer.

Output Format:

The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

```
1 n = int(input())
2
3
4 res = 0
5 while(n > 0):
6     r = n % 10
7     res = res * 10 + r
8     n = int(n / 10)
9 print(res)
10
11
```

Average time0.017 s17.00 ms

Maximum time0.027 s27.00 ms

2 out of 2 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 117 ms

Expected output5367

Actual output5367

7635

Test case 227 ms

Terminal Test cases

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1.1.5 Multiplication Table:

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1.1.5. Multiplication Table00/00

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

Print the multiplication table for the given number in the format:

```
<number> x <multiplier> = <result>
```

Note:

The character "x" is used in the output to represent multiplication.

Refer to the visible test-cases for more clarity.

Sample Test Cases

multiplica...

```
1 a = int(input())
2
3 for i in range(1,11):
4     print(a,"x",i,"=",a*i)
5
6
7
```

Average time0.019 s19.50 ms

Maximum time0.027 s27.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 117 ms

Expected output8

Actual output8

8 x 1 = 8

8 x 2 = 16

8 x 3 = 24

8 x 4 = 32

8 x 5 = 40

8 x 6 = 48

8 x 7 = 56

8 x 8 = 64

8 x 9 = 72

8 x 10 = 80

Test case 227 ms

Terminal Test cases

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1.2.1 Pass or Fail:

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1.2.1. Pass or Fail

55.07

Write a Python program that accepts the number of courses and the marks of a student in those courses.
The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

Sample Test Cases

passorFa...

```
1 a = int(input())
2 marks = list(map(int, input().split()))
3
4 if any (m<40 for m in marks):
5     print("Fail")
6 else:
7     avg = sum(marks)/ a
8     print("Aggregate Percentage: %.2f"%avg)
9     if avg>75:
10        print("Grade: Distinction")
11    elif 60<= avg <75:
12        print("Grade: First Division")
13    elif 50<= avg <60:
14        print("Grade: Second Division")
15    elif 40<= avg <50:
16        print("Grade: Third Division")
```

Average time
0.018 s
17.90 ms

Maximum time
0.031 s
31.00 ms

5 out of 5 shown test case(s) passed

5 out of 5 hidden test case(s) passed

Test case 1 31 ms

Expected output
4
55 65 78 89
Aggregate Percentage: 71.75
Grade: First Division

Actual output
4
55 65 78 89
Aggregate Percentage: 71.75
Grade: First Division

Terminal

Test cases

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1.2.2 Fibonacci Series:

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1.2.2. Fibonacci Series

18.35

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.
Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci...

```
1 def fibonacci(n):
2     if (n<=1):
3         return 0
4     else:
5         return fibonacci(n-1) + fibonacci(n-2)
6
7 n = int(input())
8 for i in range(1, n + 1):
9     print(fibonacci(i), end=" ")
10
11
12
13 def fibonacci(n):
14     if n == 0:
15         return 0
16     elif n == 1:
```

Average time
0.079 s
79.00 ms

Maximum time
0.244 s
244.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 17 ms

Expected output
5
0 1 1 2 3

Actual output
5
0 1 1 2 3

Test case 2 16 ms

Terminal

Test cases

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1.2.3 Pattern-1:

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1.2.3. Pattern - 105:26

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

rightangl...

```
1 a=int(input())
2
3 for i in range (1,a+1):
4     for j in range (1,i+1):
5         print("*",end="")
6     print()
```

Average time0.015 s15.50 msMaximum time0.019 s19.00 ms

2 out of 2 shown test case(s) passed4 out of 4 hidden test case(s) passed

Test case 119 msDebug

Expected output

Actual output

5

*
* *
* * *
* * * *
* * * * *

*
* *
* * *
* * * *
* * * * *

TerminalTest cases

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1.2.4 Pattern - 2 :

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1.2.4. Pattern - 202:59

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

- Refer to the displayed test cases for the sample pattern.

Sample Test Cases

numberP...

```
1 a=int(input())
2
3 for i in range (1,a+1):
4     for j in range(1,i+1):
5         print(j,"",end="")
6     print()
7
8
```

Average time0.017 s17.00 msMaximum time0.025 s25.00 ms

2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 118 msDebug

Expected output

Actual output

5

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

TerminalTest cases

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